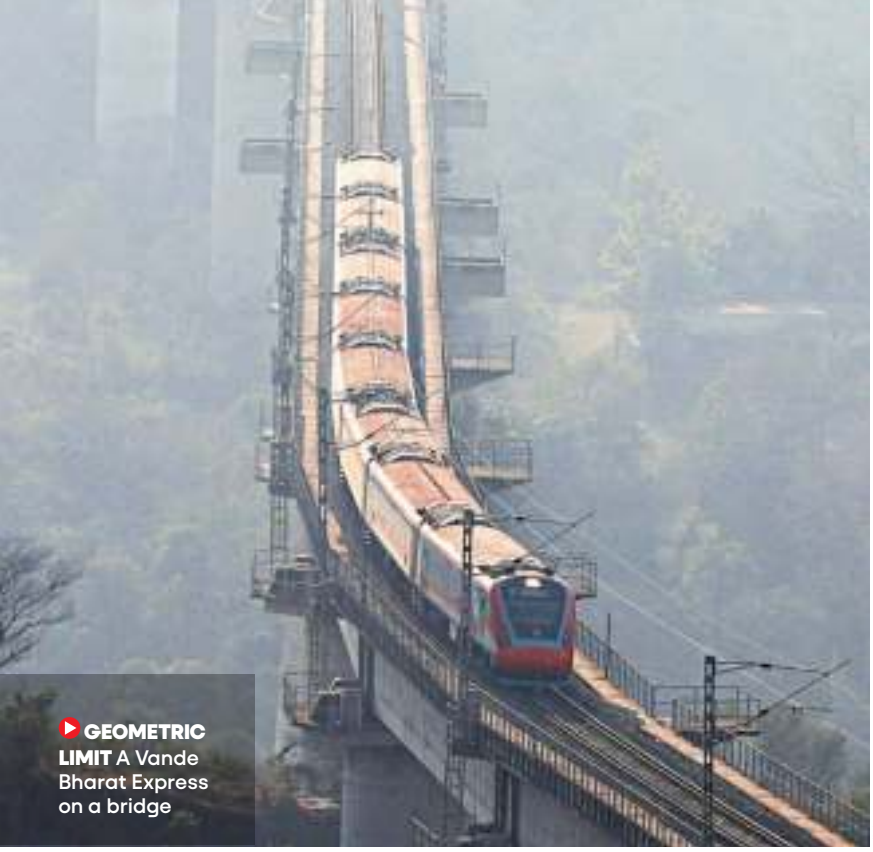


GEOMETRIC LIMIT A Vande Bharat Express on a bridge

AP

Integral Coach Factory that designed India's first Vande Bharat trainset.

Former Railway Board member (engineering) Subodh Jain points to deeper flaws. "It is misleading to say averages are only because of speed restrictions," he says. "The real issue is track condition and the way recovery time is built into schedules. Stoppages for food, water, crew changes, brake checks stretch beyond what's needed. Add congestion near cities, and the delays multiply." According to him, Mission Raftaar, launched to push up speeds, failed because it became a numbers exercise. "You built third lines, but without cutting curves or improving geometry. You laid tracks, but didn't enable trains to run faster. It was about showing kilometres built, not delivering speed."

The pattern is not new. The Rajdhani Express was hailed as a revolution when introduced in 1969. By the 1980s, it was slowed by the same track limits. "The Delhi-Howrah Rajdhani started 57 years ago at 17 hours 20 minutes," Jamshed points out. "It still takes about 17 hours 10 minutes." Ditto with the Shatabdis that came in 1988. Promising a fast, business-class ride, they, too, were slowed down within a decade.

Where Indian Railways has moved faster is in rolling out new trainsets.

Curves, gradients, bridges—most routes were never engineered for higher speeds. Despite years of announcements, not one corridor is fit for 160 kmph end to end

Up next: the Vande Bharat Sleeper. A prototype touched 180 kmph in January 2025, with 200 sets planned for overnight journeys. Subhranshu calls it ironic: "Nowhere in the world have trainsets been run in sleeper mode—not in China, Europe or Japan. The whole point of a trainset is higher speed and shorter journeys of 6-8 hours, where a chair car is enough. At 12 hours, like Delhi-Mumbai today, you're not really getting the speed advantage. Why spend Rs 120 crore on a sleeper when a conventional Rs 60 crore LHB [German coach] train can do the same job?"

The sleeper sets are undeniably deluxe—cream-and-wood interiors, cushioned berths wired for USB charging and reading lights; even hot showers in First AC, airlocked gangways, noise-attenuating insulation, WiFi and CCTV. They claim to offer more comfort than Rajdhanis. But unless the tracks catch up, they risk being luxury hotels on wheels—lovely and costly, but royally slow.

Tracks before Trains

Railways minister Ashwini Vaishnaw, who has been engaged in resolving this long-standing problem, often faces questions about Vande Bharat's speed. His explanation? "The average speed of train services, including Vande Bharat, depends on the Maximum Permissible Speed (MPS) of the section, track structure and geometry along the route, including gradients and curves, as well as topographical conditions, number of stoppages en route, line capacity utilisation, maintenance works etc." The minister points to another aspect that often goes unnoticed. "In India, we face the unique challenge of keeping train operations running while we carry out track upgrade work."

A 2024 PwC report highlights the disconnect between procurement and readiness: "The performance of Vande Bharat trainsets is impacted by the unavailability of compatible track." In other words, India bought the Teslas before fixing its highways. But Vaishnaw says, "Continuous improvement is the approach. We've gone from less than 40 per cent of tracks fit for 110 kmph in 2014 to more than 78 per cent today." Meanwhile, passengers are torn. Occupancy is strong, over 100 per cent. But online forums often bristle with complaints: "I reached in the same time as the Shatabdi. Where's the speed?"

Abroad, the lesson is clear (see *How the World Moves*). Britain runs 200 kmph Intercity expresses on tracks older than independent India. The trick was to fix the system first: fencing, signalling, grade separation. The trains themselves were not speed demons; it was the ecosystem that made speed possible.

In India, the Vande Bharat Express risks becoming a symbol of squandered opportunity unless the government invests in tracks, signalling and systemic upgrades with the same intensity it has shown for rolling stock. The question isn't whether the train can run fast—it's whether Indian Railways will ever let it. ■